


46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

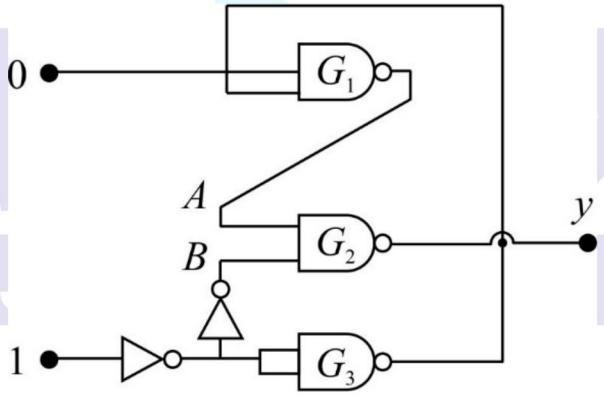
**Sec: SR STAR - A & APEX, ALL IITS** **DT: 06 / 01 /2026**
**Time: 3 Hrs** **GTM - 10** **Max. Marks: 300**

**PHYSICS** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

1. In circuit in following fig, the value of Y is




A) 0

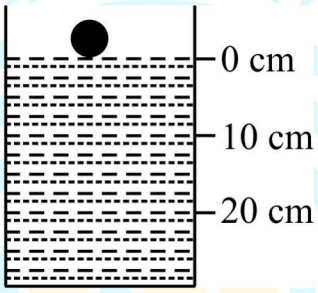
B) 1

C) fluctuates between 0 and 1

D) indeterminate as a circuit cannot be realized

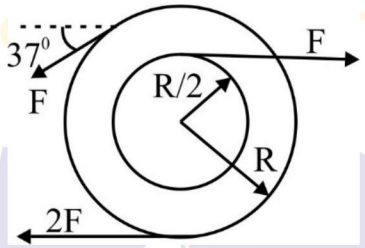
For More Join: @IITJEE_Advanc

2. A spherical ball of density $\rho$ and radius 0.003 m is dropped into a tube containing a viscous fluid filled up to the 0 cm mark as shown in the figure. Viscosity of the fluid = 1.260 $Nm^{-2}$ and its density $\rho_L = \rho/2 = 1260\ kgm^{-3}$. Assume the ball reaches a terminal speed by the 10 cm mark. The time taken (in s) by the ball to traverse the distance between the 10 cm and 20 cm mark is (g = acceleration due to gravity = 10 $ms^{-2}$)



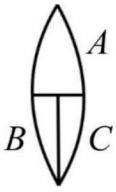

A) 2 sec B) 10 sec C) 5 sec D) 20 sec

3. On a disc of radius R a concentric circle of radius R/2 is drawn. The disc is free to rotate about a frictionless axis through its center and perpendicular to plane of the disc. All three forces (in plane of the disc) shown in figure are exerted tangent to their respective circular periphery. The magnitude of the net torque (about centre of disc) acting on the disc is




A) 1.5 FR B) 1.9 FR C) 2.3 FR D) 2.5 FR

4. A thin, symmetric bi-convex lens of power P is cut into three parts A,B and C as shown. The power of A and B is

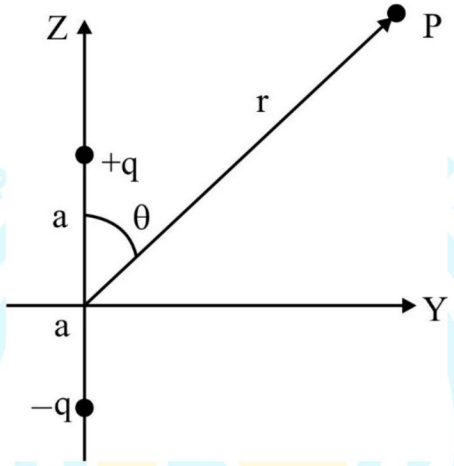





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A) $P, \frac{P}{2}$ B) $\frac{P}{2}, \frac{P}{4}$ C) $\frac{P}{4}, P$ D) $\frac{P}{2}, 2P$

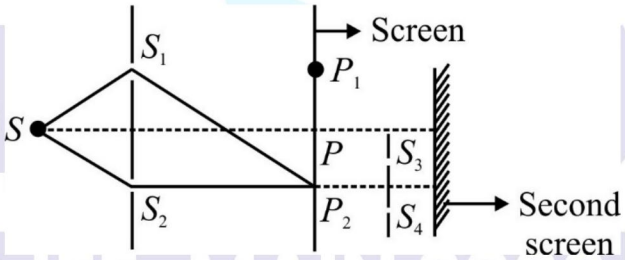
5. Figure shows an electric dipole of dipole moment $p = q(2a)$. For r>>a, what is the potential of the dipole at a point P with coordinates (x,y,z)




A) $\frac{p}{4\pi\varepsilon_0}\frac{x}{(x^2+y^2+z^2)^{3/2}}$ B) $\frac{p}{4\pi\varepsilon_0}\frac{y}{(x^2+y^2+z^2)^{3/2}}$

C) $\frac{p}{4\pi\varepsilon_0}\frac{z}{(x^2+y^2+z^2)^{3/2}}$ D) $\frac{p}{4\pi\varepsilon_0}\frac{xyz}{(x^2+y^2+z^2)^{3/2}}$

6. Figure shows a standard two slit arrangement with slits $S_1, S_2$. $P_1$ and $P_2$ are the two minima points on either side of $P$.




At $P_2$ on the screen, there is a hole and behind $P_2$ is a second 2-slit arrangement with slits $S_3, S_4$ and a second screen behind them.

A) There would be no interference pattern on the second screen but it would be lighted.

B) The second screen would be totally dark.

C) There would be a single bright point on the second screen.



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D) There would be a regular two slit pattern on the second screen.

7. Some of the thermodynamic parameters are state variables while some are process variables. Some grouping of the parameters are given. Choose the correct one

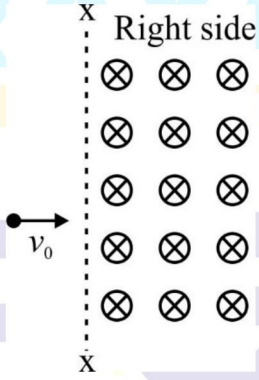
A) **state variables**: temperature, no. of moles, **process variables:** internal energy, work done by gas

B) **state variables:** volume, temperature, **process variables:** internal energy, work done by gas

C) **state variables:** work done by gas, heat rejected by gas, **process variables:** temperature, volume

D) **state variables:** internal energy, volume, **process variables:** work done by gas, heat absorbed by gas

8. A charged particle having charge $q_0$ and mass m is enters into a magnetic field with velocity $v_0$ as shown. Uniform magnetic field $B_0$ is present only on the right side of dotted line. Average force exerted on charge particle during time it is inside magnetic field is




A) $\frac{2qB_0v_0}{\pi}$ B) $qB_0v_0$ C) zero D) $2qB_0v_0$

9. statement 1: When a thin transparent sheet is placed in front of both the slits of Youngs experiment, the fringe width will increase

Statement 2 : In Youngs experiment, the fringe width is proportional to wavelength of the source used

A) statement 1 is true, statement 2 is true and it is correct explanation for statement 1



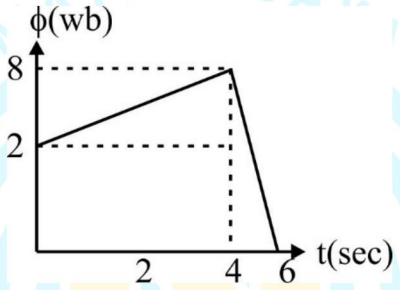
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B) statement 1 is true, statement 2 is true and but it is NOT correct explanation for statement 1

C) statement 1 is true, statement 2 is false

D) statement 1 is false, statement 2 is true

10. The magnetic flux ($\phi$) versus time (t) graph for a coil is shown in the figure. The magnitude of average emf induced in the coil for a time interval of 0 s to 6 s is

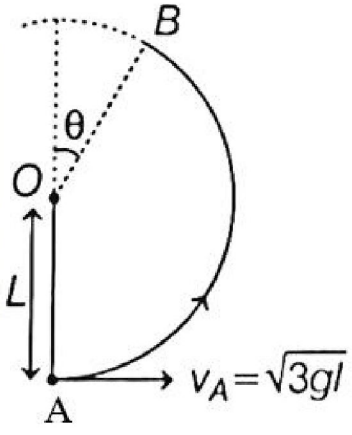



A) 1V B) 1/3V C) 2/3V D) 2V

11. A ball is thrown at an angle $\theta$ with the horizontal. Its initial kinetic energy is 100 J and it becomes 75 J at half the maximum height. The angle of projection is

A) 45° B) 30° C) $\cos^{-1}\left(\frac{3}{4}\right)$ D) $\tan^{-1}\left(\frac{4}{3}\right)$

12. A particle is projected with velocity $\sqrt{3gL}$ at point A (lowest point of the circle) in the vertical plane. Find the maximum height above A to which the particle rises




A) $\frac{24L}{27}$ B) $\frac{30L}{27}$ C) $\frac{40L}{27}$ D) $\frac{35L}{27}$



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13. Statement 1: If simple pendulum is in a carriage which is accelerating downward and acceleration greater than acceleration due to gravity, then pendulum turns upside down and oscillates about highest point.

Statement 2: The time period of pendulum will be independent of g in above case of pendulum oscillating about highest point

A) statement 1 is true, statement 2 is true and it is correct explanation for statement 1

B) statement 1 is true, statement 2 is true and but it is NOT correct explanation for statement 1

C) statement 1 is true, statement 2 is false

D) statement 1 is false, statement 2 is true

14. In a series resonant L-C-R circuit, the voltage across R is 100 V and R=1 $k\Omega$ with C = 2 $\mu F$ . The resonant frequency $\omega$ is 200 rad/s. At resonance the voltage across L is

A) $2.5\times10^{-2}V$ B) $250\ V$ C) $40\ V$ D) $4\times10^{-3}V$

15. Assume pressure (P), length (L) and velocity (V) are fundamental quantities, then the dimensions of coefficient of viscosity $(\eta)$ is

A) $[PLV^{-1}]$ B) $[PL^{-1}V]$ C) $[P^{-2}LV^{-1}]$ D) $[PL^{-1}V^{-2}]$

16. Fountains usually seen in gardens are generated by a wide pipe with an enclosure at one end having many small holes. Consider one such fountain which is produced by a pipe of internal diameter 2 cm in which water flows at a rate 3 $ms^{-1}$. The enclosure has 100 holes each of diameter 0.05 cm. the velocity of water coming out of the holes is (in $ms^{-1}$)

A) 0.48 B) 96 C) 24 D) 48

17. In a photoelectric experiment, the wavelength of the light incident on the metal is changed from 200 nm to 400 nm. The decrease in the stopping potential is close to [use hc=1240ev-nm, h-Planck's constant c- velocity of light]

A) 3.1 V B) 2.8 V C) 4.2 V D) 1.2 V



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18. In an experiment in determination of density of a metal given in the form of a small cylindrical rod, the following observations are recorded

i) mass of cylinder = $(6.7 \pm 0.1)g$

ii) radius of cylinder = $(0.287 \pm 0.001)cm$

iii) length of cylinder = $(3.28 \pm 0.01)cm$

Find percentage error in measurement of density

A) 1.4% B) 3.5% C) 3.0% D) 2.5%

19. A polytropic process for an ideal gas is represented by equation $pV^n$=constant. If $\gamma$ is ratio of specific heats $\left(\frac{C_p}{C_v}\right)$, then value of n for which molar heat capacity of the process is negative, is given as

A) $\gamma$>n B) $\gamma$>n>1

C) n> $\gamma$ D) none, as it is not possible

20. A particle moves along the y-axis of a coordinate system, with a force component $F_y = (2N/m^3)y^3$ acting on it. As the particle moves from the origin to y=3m, how much work is done on it by the force?

A) –162 J B) –40.5 J C) 40.5 J D) 162 J

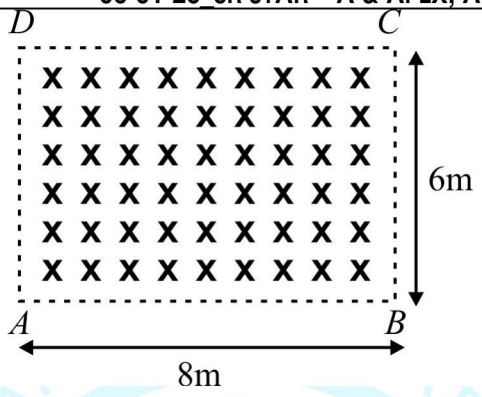
## SECTION – II
## (Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

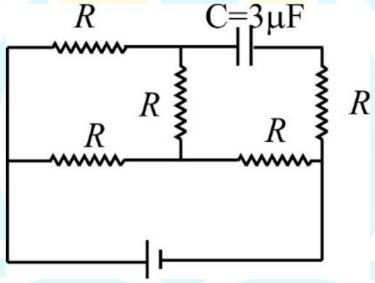
21. A charged particle of mass m = 5 g and charge $q = 12\mu C$ enters along AB at a point A in a uniform magnetic field of magnitude B = 100 T in an existing rectangular region in size (8m x 6m) and into the plane of paper as shown in figure, the particle leaves the region at C. The speed of the charged particle (in m/s) as it leaves the region is



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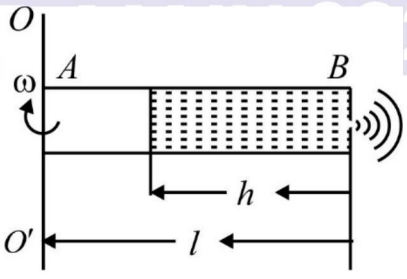



22. In the given circuit, the potential difference across the capacitor is 12V in steady state. Each resistor have $3\Omega$ resistance. The emf of the ideal battery is




23. A parallel plate capacitor consists of two circular plates each of radius 2cm, separated by a distance of 0.1 mm. If voltage across the plates is varying at the rate of $5\times10^{13}V/s$, then the value of displacement current is N x $10^3$, the value of N is

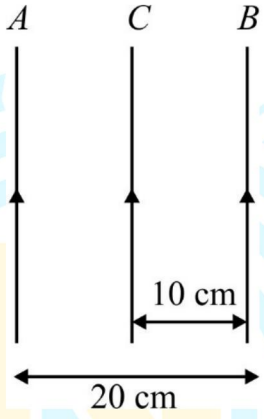
24. A horizontally oriented tube AB of length $l$ rotates with a constant angular velocity $\omega$ about a stationary vertical axis $OO'$ passing through the end A. The tube is filled with an ideal fluid. The end A of the tube is open; the closed end B has a very small orifice, velocity of the fluid relative to the tube as a function of the column height h is $v=\omega h\sqrt{\frac{Nl}{h}-1}$, the value of N is






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25. In the adjoining figure, two very long parallel wires A and B carry currents of 10A and 20A, respectively and are at a distance 20 cm apart. If a third wire C (length 15 cm) having a current of 10A is placed between them, then the magnitude of net force will act on C is $N\times10^{-5}$ then the value of N is




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**CHEMISTRY** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

26. **Statement-I:** If nitrogen and sulphur both are present in an organic compound. It gives blood red colour in Lassaigne's test [avoid excess of sodium]

**Statement-II:** In Lassaigne's test, sulphur present in an organic compound is converted into sodium sulphate by fusing the compound with sodium metal

(A) Both Statement-I and Statement-II are true

(B) Both Statement-I and Statement-II are false

(C) Statement-I is true but Statement-II is false

(D) Statement-I is false but Statement-II is true

27. 2L of 0.1 M $H_2SO_4$ is allowed to react with 2L of 0.1 M NaOH. Select the correct statement

(A) Molarity of $Na_2SO_4$ formed is 0.1 M

(B) Molarity of $H_2SO_4$ left is 0.1 M

(C) Mass of $Na_2SO_4$ formed is 14.2 g

(D) pH of the resultant solution = 2

28. Which of the following is incorrect statement?

(A) The order of wavelength of light absorbed by the complexes:

$[Ni(H_2O)_2(en)_2]^{2+} < [Ni(H_2O)_4(en)]^{2+} < [Ni(H_2O)_6]^{2+}$

(B) The order of crystal field stabilization energies of complexes:

$[Co(NH_3)_6]^{3+} > [Rh(NH_3)_6]^{3+} > [Ir(NH_3)_6]^{3+}$

(C) The energy of $t_{2g}$ orbitals is more than the energy of $e_g$ orbitals in tetrahedral coordination entities

(D) The order of crystal field stabilization energies of complexes:

$[Co(SCN)_6]^{3-} < [Co(H_2O)_6]^{3+} < [Co(NCS)_6]^{3-} < [Co(CN)_6]^{3-}$



For More Join: @IITJEE_Advanc

29. **Statement-I:** Zn corrodes in 2M acid solution but not in aq. $Na_2SO_4$ solution.

**Statement-II:** Corrosion can be prevented by physically separating the metal from the environment using protective coatings like paint or grease

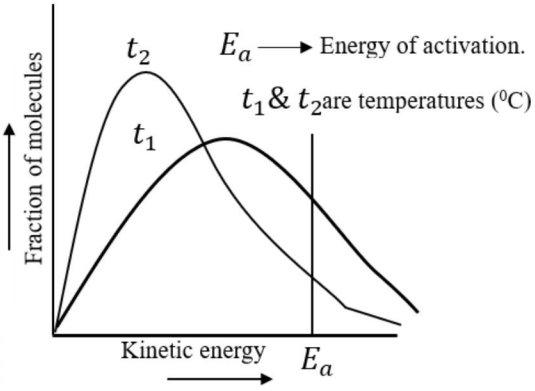
(A) Both Statement-I and Statement-II are true

(B) Both Statement-I and Statement-II are false

(C) Statement-I is true but Statement-II is false

(D) Statement-I is false but Statement-II is true

30. The incorrect statement for below distribution curve



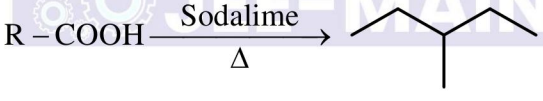

(A) $t_1 > t_2$

(B) The fraction of molecules which react at $t_2$ temperature is more than the fraction of molecules which react at $t_1$ temperature

(C) The area under the curve must be same at $t_1$ and $t_2$ temperatures

(D) When the temperature is raised the maximum of the curve moves to the higher kinetic energy value

31. The maximum number of different carboxylic acids (RCOOH) that can give 3-methyl pentane by the following reaction is _____ (including stereo isomers)




(A) 2 (B) 4 (C) 6 (D) 8



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32. Which of the following is not disproportionation reaction?

(A) $2H_2O_2 \longrightarrow 2H_2O + O_2$

(B) $P_4 + 3OH^- + 3H_2O \longrightarrow PH_3 + 3H_2PO_2^-$

(C) $2F_2 + 2OH^- \longrightarrow 2F^- + OF_2 + H_2O$

(D) $Cl_2 + 2OH^- \longrightarrow ClO^- + Cl^- + H_2O$

33. The solubility of $Pb(OH)_2$ in water is $6.7\times10^{-6}$ M. Then the solubility of $Pb(OH)_2$ in a buffer solution of pH = 8 is

(A) $1.203\times10^{-3}$ M (B) $3.42\times10^{-6}$ M

(C) $8.163\times10^{-4}$ M (D) $4.208\times10^{-5}$ M

34. Which of the following is/are correct statements?

(I) $[Co(NH_3)_6]^{3+}$, $[Ni(CN)_4]^{2-}$ and $[Co(C_2O_4)_3]^{3-}$ are inner orbital complexes

(II) $[Mn(CN)_6]^{3-}$, $[Fe(CN)_6]^{3-}$ and $[CoF_6]^{3-}$ are paramagnetic complexes

(III) $[Ni(CN)_4]^{2-}$, $[Ni(CO)_4]$ and $[PtCl_4]^{2-}$ are square planar complexes

(IV) All complexes of $Ti^{3+}$, $V^{3+}$ and $Cr^{3+}$ are paramagnetic

(A) I, II, III and IV (B) I, III and IV only

(C) I, II and IV only (D) I, II only

35. Match the List-I with List-II

| List-I | | List-II | |
|---|---|---|---|
| (P) | Isothermal change | (1) | $P_{ext} = 0$ |
| (Q) | Adiabatic process | (2) | At constant volume |
| (R) | $\Delta U = q$ | (3) | No exchange of heat |
| (S) | Free expansion | (4) | Constant temperature |

The correct match is

(A) P-(3); Q-(4); R-(2); S-(1) (B) P-(4); Q-(3); R-(1); S-(2)

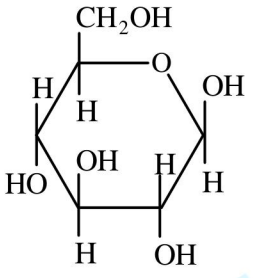
(C) P-(2); Q-(3); R-(4); S-(1) (D) P-(4); Q-(3); R-(2); S-(1)



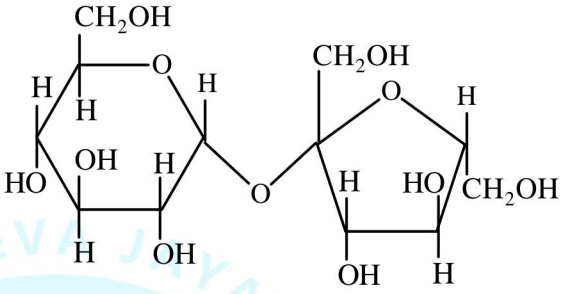
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36. How many of the following are reducing sugars

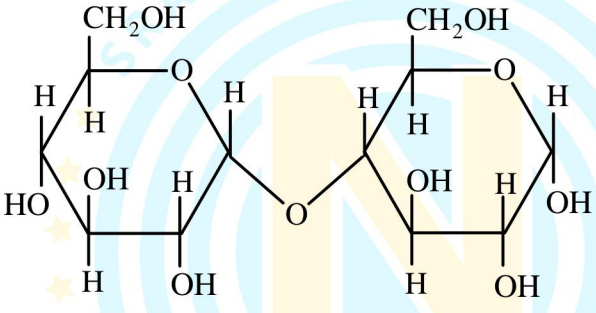
(i)



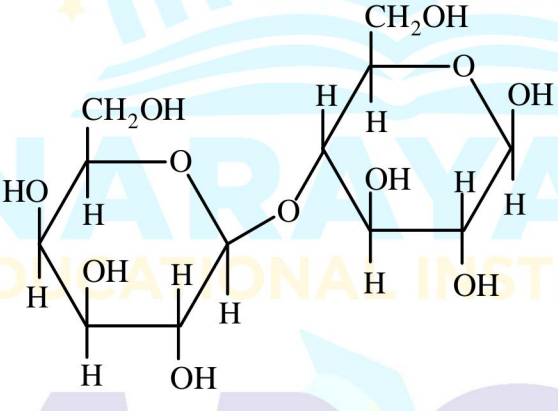
(ii)



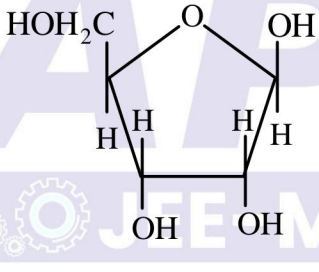
(iii)



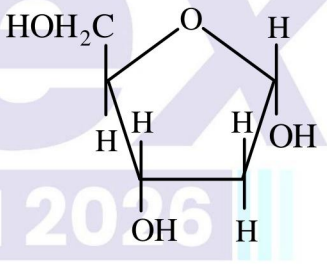
(iv)



(v)



(vi)



(A) 6 (B) 5 (C) 4 (D) 3



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37. **Statement-I:** The wavelength of any line in Lyman series of hydrogen spectrum is lower than the wavelength of series limit of Balmer series of hydrogen spectrum

**Statement-II:** The series limit of Balmer series of hydrogen spectrum appears in violet region.

(A) Both Statement-I and Statement-II are false

(B) Both Statement-I and Statement-II are true

(C) Statement-I is true but Statement-II is false

(D) Statement-I is false but Statement-II is true

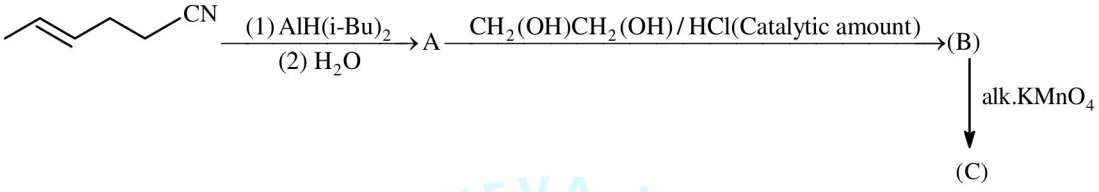
38. Choose the correct option in which both Statement I and Statement II are correct and Statement II is not the correct explanation for Statement I

| | **Statement I** | **Statement II** |
|---|---|---|
| A) | Certain chemical properties of Tl (group 13) are similar to that of Na (group 1) | Inert pair effect |
| B) | Atomic size of Zr is almost same as that of Hf | Lanthanide contraction |
| C) | Certain properties of Be (group 2) are similar to that of Al (group 13) | Polarizing power of cation |
| D) | $CO_2$ is acidic and $SnO_2$ is amphoteric oxide | Both C and Sn belong to same group (group 14) |



For More Join: @IITJEE_Advanc

39. The number of oxygen atoms in a molecule of C is




(A) 2 (B) 3 (C) 4 (D) 6

40. The structure of 4 – methylpent – 2 – en – 1 – ol is_____

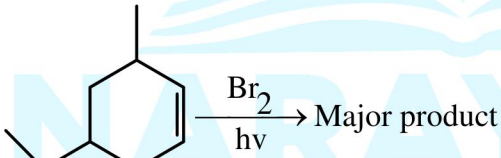
A) $CH_3CH_2CH=CHCH_2OH$

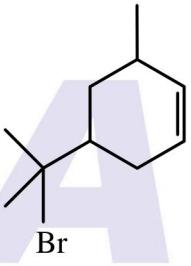
B) $(CH_3)_2C=CHCH_2CH_2OH$

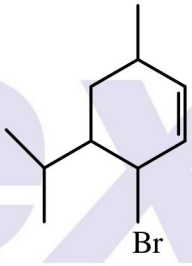
C) $(CH_3)_2CHCH=CHCH_2OH$

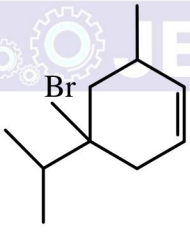
D) $CH_3CH(OH)CHCH=C(CH_3)_2$

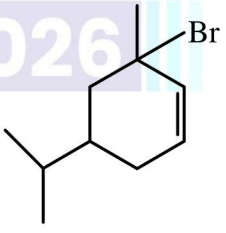
41. The major product of the following reaction is




(A) 


(B) 

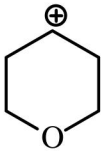

(C) 

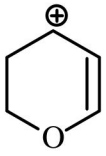

(D) 

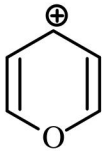


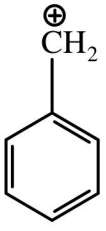

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42. The most stable carbocation from the following is

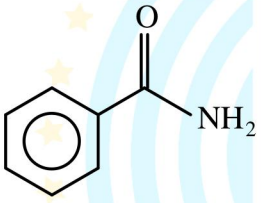
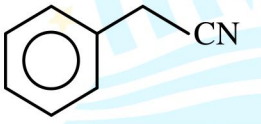
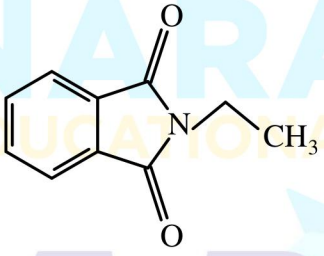
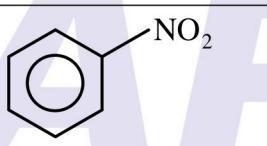
(A) 

(B) 

(C) 

(D) 

43. Match the compounds (List-I) with the appropriate reagents (List-II) for their conversion into corresponding amines.

| List-I | | List-II | |
|---|---|---|---|
| (P) |  | (1) | $Fe + HCl$ |
| (Q) |  | (2) | $Br_2$ / alc. NaOH |
| (R) |  | (3) | aq. NaOH |
| (S) |  | (4) | $LiAlH_4 / H_2O$ |

The correct match is

(A) P-(3); Q-(1); R-(2); S-(4)

(B) P-(2); Q-(4); R-(3); S-(1)

(C) P-(2); Q-(3); R-(4); S-(1)

(D) P-(4); Q-(1); R-(2); S-(3)



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44. Arrange the following compounds in increasing order of their dipole moment:

HF, $H_2O$, $NH_3$ and HBr

(A) $H_2O > HF > NH_3 > HBr$ (B) $HF > H_2O > NH_3 > HBr$

(C) $HF > H_2O > HBr > NH_3$ (D) $H_2O > HF > HBr > NH_3$

45. Which of the following parameter is not useful to calculate standard electrode potentials of halogens?

(A) Bond dissociation enthalpy of halogen $(X_2)$

(B) Electron gain enthalpy of halogen

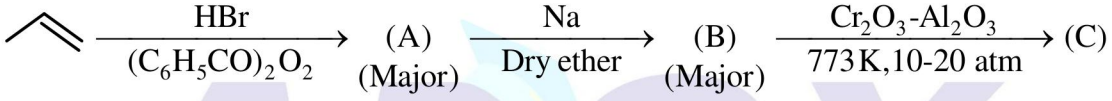
(C) Ionization enthalpy of halogen

(D) Hydration enthalpy of halide ion $(X^-)$

## SECTION – II
## (Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

46. The percentage of carbon (by weight) in the product 'C' in the following reaction sequence is $P \times 10^{-1}$, the value of P is [Nearest Integer]




47. The volume (in mL) of 0.1M aqueous KOH solution required to react completely with 20 mL of 0.1M aqueous solution of $H_3PO_3$ is___

48. The enthalpy changes of the following reactions at $27^0C$ are

$Na(s) + \frac{1}{2}Cl_2(g) \longrightarrow NaCl(s)$ ; $\Delta H_1^{\ominus} = -411$ kJ / mole

$H_2(g) + S(s) + 2O_2(g) \longrightarrow H_2SO_4(\ell)$ ; $\Delta H_2^{\ominus} = -811$ kJ / mole

$2Na(s) + S(s) + 2O_2(g) \longrightarrow Na_2SO_4(s)$ ; $\Delta H_3^{\ominus} = -1382$ kJ / mole



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$\frac{1}{2}H_2(g)+\frac{1}{2}Cl_2(g)\longrightarrow HCl(g)$; $\Delta H_4^{\ominus}=-92$ kJ / mole

From these data, the enthalpy change of reaction (in kJ) at $27^0C$ for the process $2NaCl(s)+H_2SO_4(\ell)\longrightarrow Na_2SO_4(s)+2HCl(g)$ is _____

49. How many of the following are correct statements?

(i) $CrO_4^{2-}$ is converted to $Cr_2O_7^{2-}$ in acidic aqueous solution.

(ii) In $Cr_2O_7^{2-}$ ion, $Cr-O$ (formal charge of oxygen is zero) bond length is more than $Cr-O$ (formal charge of oxygen is not zero) bond length.

(iii) The manganate and permanganate ions are tetrahedral

(iv) $Mn-O$ bond length in manganate ion is more than $Mn-O$ bond length in permanganate ion

50. A mixture of $Zn^{2+}$, $Mn^{2+}$, $Ni^{2+}$ and $Co^{2+}$ ions on reaction with $NH_4Cl+NH_4OH+H_2S$ gas gives 'x' number of metal sulphide precipitates and on further reaction with conc. HCl 'y' number of metal sulphide precipitates are soluble. The value of x + y is_____



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**MATHEMATICS** **MAX.MARKS: 100**

## SECTION – I
## (SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

51. Let matrix $A=\begin{bmatrix} x & y & -z \\ 1 & 2 & 3 \\ 1 & 1 & 2 \end{bmatrix}$ where $x, y, z \in N$. If $\det(\text{adj}(\text{adj}\, A)) = 2^8 \cdot 3^4$, then the number of such matrices A is [Note: adj A denotes adjoint of square matrix A]

(a) 220 (b) 45 (c) 55 (d) 110

52. The number of ways in which 6 gentlemen and 3 ladies be seated round a table so that every gentleman may have a lady by his side is

(a) 1440 (b) 720 (c) 240 (d) 480

53. Let $(1+x^2)^2(1+x)^n = A_0 + A_1x + A_2x^2 + \ldots..$ If $A_0, A_1, A_2$ are in A.P., then the possible value of $'n'$ is

(a) 9 (b) 3 (c) 5 (d) 7

54. Let $P(\alpha, \beta, \gamma)$ be the foot of the perpendicular from the point $A(3,-1,4)$ on the line $L: \frac{x-2}{2} = \frac{y-3}{-3} = \frac{z+1}{1}$, then $\alpha+\beta+\gamma$ equals :

(a) 2 (b) 4 (c) 6 (d) 8

55. For $P(A)=\frac{3}{8};\ P(B)=\frac{1}{2};\ P(A\cup B)=\frac{5}{8}$ which of the following does not hold good?

(a) $P\left(\frac{A^c}{B}\right) = 2P\left(\frac{A}{B^c}\right)$ (b) $P(B) = P\left(\frac{A}{B}\right)$

(c) $15P\left(\frac{A^c}{B^c}\right) = 8P\left(\frac{B}{A^c}\right)$ (d) $P\left(\frac{A}{B^c}\right) = P(A\cap B)$



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56. $\int \frac{e^{\tan^{-1}x}}{(1+x^2)}\left[\left(\sec^{-1}\sqrt{1+x^2}\right)^2+\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)\right]dx \ (x>0) =$

(a) $e^{\tan^{-1}x}\cdot\tan^{-1}x+C$

(b) $\frac{e^{\tan^{-1}x}\cdot(\tan^{-1}x)^2}{2}+C$

(c) $e^{\tan^{-1}x}\cdot(\sec^{-1}(\sqrt{1+x^2}))^2+C$

(d) $e^{\tan^{-1}x}\cdot(\cos ec^{-1}(\sqrt{1+x^2}))^2+C$

57. The area bounded by the curve $y^2\le 4x$ and the circle $x^2+y^2-2x-3\le 0$ is equal to

(a) $2\pi+\frac{8}{3}$ (b) $4\pi-\frac{8}{3}$ (c) $\pi+\frac{8}{3}$ (d) $\pi-\frac{8}{3}$

58. For $x\in\left(0,\frac{5\pi}{2}\right)$, if $f(x)=\int_0^x\sqrt{t}\sin t dt$, then $f(x)$ has

(a) Local minimum at $\pi$ and local maximum at $2\pi$

(b) Local maximum at $\pi$ and Local minimum at $2\pi$

(c) Local maximum at $x=\pi,2\pi$

(d) Local minimum at $x=\pi,2\pi$

59. PSQ is a focal chord of a parabola $y^2=4ax$ whose focus is S and vertex is A. PA and QA are produced to meet the directrix in R and T, respectively. Then $\angle RST$ =

(a) $30^0$ (b) $90^0$ (c) $60^0$ (d) $45^0$

60. $\vec{a}$ and $\vec{b}$ are two unit vectors inclined at an angle $\alpha$ ($\alpha\in[0,\pi]$) to each other. If $|\vec{a}+\vec{b}|<1$, then $\alpha$ cannot lie in

(a) $\alpha\in\left(\frac{\pi}{3},\frac{2\pi}{3}\right)$ (b) $\alpha\in\left(\frac{2\pi}{3},\frac{3\pi}{4}\right)$

(c) $\alpha\in\left(\frac{3\pi}{4},\frac{5\pi}{6}\right)$ (d) $\alpha\in\left(\frac{5\pi}{6},\pi\right)$



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61. $\lim\limits_{x\to 0}\left(\dfrac{(1+x)^{2/x}}{e^2}\right)^{\frac{4}{\sin x}}$ is

(a) $e^4$ (b) $e^{-4}$ (c) $e^8$ (d) $e^{-8}$

62. Consider all function $f:\{1,2,3,4\}\to\{1,2,3,4\}$ which are one-one onto and satisfy the following property, if $f(k)$ is odd then $f(k+1)$ is even, $k=1,2,3$. The number of such function is

(a) 4 (b) 8 (c) 12 (d) 16

63. The number of terms of an A.P., is odd. The sum of the odd terms ($1^{st}$, $3^{rd}$ etc.,) is 248 and the sum of the even terms is 217. The last term exceeds the first by 56, then

(a) the number of terms is 17 (b) the first term is 3

(c) the number of terms is 13 (d) the first term is 1

64. Let $f(x)$ be differentiable function on the interval $(0,\infty)$ such that $f(1)=1$ and $\lim\limits_{t\to x}\left(\dfrac{t^3f(x)-x^3f(t)}{t^2-x^2}\right)=\dfrac{1}{2}\forall x>0$, then $f(x)$ is

(a) $\dfrac{1}{4x}+\dfrac{3x^2}{4}$ (b) $\dfrac{3}{4x}+\dfrac{x^3}{4}$ (c) $\dfrac{1}{4x}+\dfrac{3x^3}{4}$ (d) $\dfrac{1}{4x^3}+\dfrac{3x}{4}$

65. The system of equations $x+y+z=6$, $x+2y+3z=10$, $x+2y+\lambda z=k$ is inconsistent if $\lambda$ = ……., $k\neq$ ……

(a) 3, 7 (b) 3, 10 (c) 7, 10 (d) 10, 3

66. Let $y=f(x)$ and $\dfrac{x}{y}\dfrac{dy}{dx}=\dfrac{3x^2-y}{2y-x^2}$; $f(1)=1$ then the possible value of $\dfrac{1}{3}f(3)$ equals

(a) 9 (b) 4 (c) 3 (d) 2

67. Let $f(x)=x^2+ax+b$ where $a,b\in R$. If the maximum and the minimum values of $f(x)$ are 3 and 2 respectively for $0\le x\le 2$, then a possible ordered pair $(a,b)$ is

(a) $(-2,3)$ (b) $\left(-\dfrac{3}{2},2\right)$ (c) $\left(-\dfrac{5}{2},3\right)$ (d) $\left(-\dfrac{5}{2},2\right)$



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68. The number of solutions in $[0,2\pi]$ satisfying the equation $(\log_{\sqrt{3}}\tan x)\left(\sqrt{(\log_{\tan x}3+\log_{\sqrt{3}}3\sqrt{3})}\right)=-1$ is

(a) 0 (b) 2 (c) 3 (d) > 4

69. Let a complex number '$z$', with minimum argument is lying on the curve $|z+4i|=2$ then $\left|\arg(z)+\sum_{r=0}^{17}\left(\frac{z}{2\sqrt{3}}\right)^r\right|=$

(a) $5-\frac{\pi}{3}$ (b) $3-\frac{\pi}{3}$ (c) $\frac{\pi}{3}$ (d) $\frac{2\pi}{3}$

70. PQR is an equilateral triangle such that the vertices Q and R lies on the lines $x+y=\sqrt{2}$ and $x+y=7\sqrt{2}$ respectively. If P lies between the two lines at distance of 4 from one of them, then the length of side of equilateral triangle PQR is (in units)

(a) 8 (b) $\frac{4\sqrt{7}}{\sqrt{3}}$ (c) $\frac{\sqrt{85}}{3}$ (d) $\frac{4\sqrt{5}}{\sqrt{3}}$

## SECTION – II
## (Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
**Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.**

71. Let $E:\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a<b)$ and $H:\frac{-x^2}{A^2}+\frac{y^2}{B^2}=1$. Let the distance between the foci of E and the foci of H be $2\sqrt{5}$ each. If the ratio of the eccentricities of E and H is $\frac{1}{2}$ and $b+B=6$, then twice the difference of the lengths of their latus rectums is equal to .

72. If $A=\{1,2,3\}$, then the number of reflexive relations in '$A$' is _____

73. Let $(p,q),(a\cos\theta,b\sin\theta),(b\cos\theta,a\sin\theta)$ are the vertices of a triangle, where '$\theta$' is parameter. If the locus of centroid of the triangle is $(\alpha x-p)^2+(\beta y-q)^2=(a+b)^2$, then $\alpha^2+\beta^2=$



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74. Given $\sum_{r=0}^{n}\left(\frac{r+3}{r+1}\right){}^{n}C_{r}=\frac{7\cdot 2^{10}-2}{10}$ then value of '$n$' is _____

75. The perpendicular distance of the point $(2,4,-1)$ from line $\frac{x+5}{1}=\frac{y+3}{4}=\frac{z-6}{-9}$ is _____

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